

Exam Date & Time: 22-Nov-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech VII Semester - End Semester Examination - Nov / Dec -2024

ARTIFICIAL INTELLIGENCE: SEARCH METHODS FOR PROBLEM SOLVING [NPT-4010]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain the emergence of MIND in philosophy with neat diagram with respect to the European medieval world view. (4)
 - A)
 - B) Explain the three laws of robotics. (4)
 - C) What is the motto of Alan Turing's Imitation Game? (2)
- 2) Distinguish the properties of state spaces. (4)
 - A)
 - B) Analyse the algorithm for BFS (4)
 - C) How do you model many real-life problems as a state space search problem (2)

- 3)

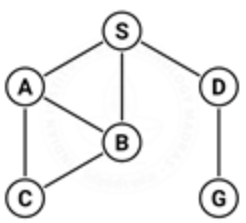
State Space

X	MoveGen(X)
S	[A, B, C]
A	[S, B, D]
B	[S, A, D]
C	[S, G]
D	[A, B, E]
E	[D, G]
G	[C, E]

(4)

Discover the search trees for the given state space.

- B) Analyze the human cognitive approach to artificial intelligence. How does it differ from the information processing view? Provide a comparative analysis with supporting examples. (4)
- C) Write the algorithm for DFID-N and DFID-C along with ancillary functions. (2)

- 4)
- Construct DFS Trees case1 and case2 for the given state space**
- A) **State Space**
- 
- | X | MoveGen(X) |
|---|------------|
| S | [A, B, D] |
| A | [C, B, S] |
| B | [S, A, C] |
| C | [B, A] |
| D | [S, G] |
| G | [D] |
- (4)
- B) Explain breadth first heuristic search (4)
- C) Outline the problems with IDA* algorithm. (2)
- 5) Explain planning domain description languages (4)
- A) (4)
- B) Explain AO* Algorithm (4)
- C) What is a "local maximum" in heuristic define? (2)

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